Civic Sense AI: Sustainable Community Issue Reporting Assistant

SDG Alignment

Primary SDG:

SDG 11 – Sustainable Cities and Communities

Secondary SDGs:

• SDG 6 – Clean Water and Sanitation

• SDG 12 – Responsible Consumption and Production

Problem Statement

Citizens frequently encounter civic issues such as garbage accumulation, water leakage, broken streetlights, potholes, and blocked drainage systems. However, many people do not know the appropriate authority to contact or how to prepare a proper complaint. As a result, these issues often remain unresolved, affecting public safety, environmental sustainability, and quality of life.

How Might We Statement

How might we use AI to help citizens identify, classify, and report community issues efficiently so that cities become cleaner, safer, and more sustainable?

Target Users

• Citizens

• Students

• Local Residents

• Community volunteers

• Resident welfare associations

• Municipal authorities

Existing Challenges

• Lack of awareness about reporting procedures.

• Difficulty identifying the responsible department.

• Poorly written complaints.

• Delayed reporting of civic issues.

• Lack of guidance regarding issue severity.

Impact of the Problem

• Water wastage

• Environmental pollution

• Public safety concerns

• Reduced community participation

Proposed AI Solution

Civic Sense AI is an AI-powered civic assistance platform that helps users report community issues effectively. Users describe an issue in natural language, and the AI system analyzes the report, identifies the issue category, determines its severity, recommends the responsible authority, and generates a professional complaint draft.

The solution also provides sustainability-related insights and recommendations based on municipal guidelines.

AI Components Used

Prompt Engineering

Users provide issue descriptions using natural language prompts.

Example:

"There is a water leakage near the bus stand for the past two days."

The AI processes the prompt and generates structured outputs.

IBM Granite Model

IBM Granite can be used for:

• Issue classification

• Complaint generation

• Information summarization

• Sustainability recommendations

Agentic AI Workflow

The system uses multiple AI agents to perform specialized tasks.

Agent 1 – Issue Classification Agent

Identifies the category of the reported issue.

Examples:

• Water Leakage

• Garbage Accumulation

• Streetlight Failure

• Road Damage

Agent 2 – Severity Analysis Agent

Determines issue priority.

Categories:

• Low

• Medium

• High

Agent 3 – Authority Finder Agent

Identifies the responsible government department.

Examples:

• Water Department

• Municipal Sanitation Department

• Electrical Maintenance Department

• Public Works Department

Agent 4 – Complaint Generation Agent

Generates a professional complaint draft for submission.

RAG (Retrieval-Augmented Generation) Component

The solution incorporates a RAG-based knowledge retrieval layer.

Knowledge Sources

• Municipal waste management guidelines

• Water conservation policies

• Road maintenance procedures

• Public safety regulations

• Civic reporting guidelines

The AI retrieves relevant information before generating recommendations and complaint summaries.

Workflow Architecture

Citizen Reports Issue

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Issue Classification Agent

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Severity Analysis Agent

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Authority Finder Agent

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RAG Knowledge Base

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Complaint Generation Agent

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Final Complaint & Guidance

Sample Use Cases

Use Case 1: Water Leakage

User Input:

"There is a water leakage near the bus stand for the past two days."

AI Output:

• Category: Water Leakage

• Severity: Medium

• Authority: Municipal Water Department

Generated Complaint:

A continuous water leakage has been observed near the bus stand for the past two days. This is causing water wastage and inconvenience to the public. Immediate inspection and repair are required.

Use Case 2: Garbage Accumulation

User Input:

"Garbage has accumulated near the government school for a week."

AI Output:

• Category: Waste Management

• Severity: High

• Authority: Municipal Sanitation Department

Generated Complaint:

A large amount of garbage has accumulated near the government school and has not been cleared for several days. This may create hygiene and health concerns. Immediate cleaning is required.

Use Case 3: Broken Streetlight

User Input:

"The streetlight near the market is not functioning."

AI Output:

• Category: Streetlight Maintenance

• Severity: Medium

• Authority: Electrical Maintenance Department

Generated Complaint:

A streetlight near the market area is not functioning properly, creating visibility and safety concerns during the nighttime. Kindly arrange for immediate repair.

Responsible AI Considerations

Fairness

The system provides equal assistance to all users without discrimination.

Transparency

The AI explains issue of classification and recommendations.

Privacy

No sensitive personal information is stored without user consent.

Human Oversight

Users review generated complaints before submission.

Ethical Use

The system is designed only for civic issue reporting and public benefit.

Expected Impact

Social Impact

• Increased civic participation

• Improved communication between citizens and authorities

• Better public safety

Environmental Impact

• Reduced water wastage

• Cleaner public spaces

• Improved waste management

Economic Impact

• Faster issue resolution

• Reduced maintenance costs through early reporting

Future Enhancements

• Direct integration with municipal complaint portals

• Mobile application support

• Image-based issue detection

• Multilingual support

• Real-time status tracking

• GPS-based issue location identification

Conclusion :

Civic Sense AI leverages Prompt Engineering, IBM Granite, Agentic AI, and RAG concepts to assist citizens in reporting civic issues efficiently. The solution promotes sustainable urban development, responsible resource management, and active community participation, contributing to SDG 11 – Sustainable Cities and Communities.